

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	1.6 m
Total 1 mark			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Uses correct forces to form a moment equation (PI)	AO1.1a	M1	Take moments about C: $Mg \times 0.8 = 0.7 \times 24$
Obtains correct value	AO1.1b	A1	$M = 21$
Total 2 marks			

Q3.

(a) (i) Moments about Q

M1

$$2.2 \times 25g = T_p \times 4.2$$

$$T_p = 13.095 \times g$$

A1

$$T_p = 128 \text{ N}$$

A1

Resolving vertically

$$T_p + T_q = 25g \text{ or } 245$$

M1

$$T_q = 117 \text{ N}$$

A1

OR

Moments about any point M1A1.

Moments about any other point M1.

$$T_P A1 ; T_Q A1$$

5

(ii) Weight of plank acts through its centre.

E1

1

(b) Resolve vertically

Could use T rather than T_P, T_Q

M1

$$T_P + T_Q = (25 + m)g = 2T_P$$

A1

Moments about B

M1

$$T_P \times 5 + T_Q \times 0.8 = 25g \times 3$$

$$(25 + m)g \times 2.9 = 25g \times 3$$

A1

$$2.9mg = 25g \times 0.1$$

$$29m = 25$$

M1

$$m = 0.862 \text{ or } \frac{25}{29}$$

A1

OR

Moments about Q

$$T_P \times 4.2 = 25g \times 2.2 - mg \times 0.8$$

$$\frac{1}{2} \times (25 + m)g \times 4.2$$

$$= 25g \times 2.2 - mg \times 0.8$$

$$2.9mg = 25g \times 0.1$$

$$29m = 25$$

OR

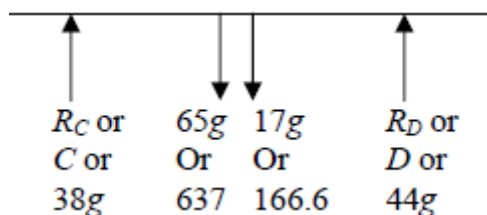
Moments about any point M1A1

Moments about any other point M1A1

Solution M1A1

Q4.

(a)



B1: Two weights correct and in correct relative positions.

B1: Two upward reaction forces, labelled differently.

Note all forces must be shown as arrows and have labels.

Condone use of $g = 9.81$ for calculating weights.

B1
B1

2

(b) Taking moments about C

$$3 \times 17g + 2.6 \times 65g = 44g \times d$$

B1: Seeing 2.6.

M1: Three term moment equation including 17g, 65g and 44g or 17, 65 and 44, with different distances for the 17g and 65g.

A1: Correct equation.

B1
M1
A1

$$44d = 220$$

$$d = 5$$

$$\text{Distance is } 5 - 4.6 = 0.4 \text{ m}$$

A1: Correct final answer.

A1

4

Alternative

$$R_C = 38g$$

(B1)

Taking moments about D

Could take moments about any other point

$$38g(4.6 + x) = 65g(2 + x) + 17g(1.6 + x)$$

(M1)
(A1)

$$174.8 - 130 - 27.2 = 44x$$

$$x = 0.4$$

(A1)

(4)

- (c) Gravitational force (centre of mass or weight) at mid-point (or centre) of the plank

E1

E1: Correct explanation.

1

[7]

Q5.

(a)



B1 for four forces

B2 for two different reactions and 30g and 20g marked

B2

2

- (b) Taking moments about A:

$$3.2 \times 30g = R_B \times 5$$

B1 for 3.2

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M1B1

$$R_B = 19.2g$$

AG

A1

3

- (c) Resolve vertically: $R_A + R_B = 50g$

Can be awarded in (b)

M1

$$R_A = 30.8g \text{ or } 302 \text{ N}$$

A1

2

- (d) Gravitational force acts through mid-point of the rod

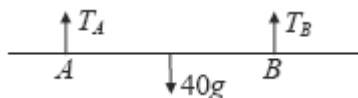
E1

1

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Q6.

(a)



B1

1

(b) Taking moments about A

$$2.1 \times 40g = T_B \times 4$$

B1 for 2.1

M1 B1

$$T_B = 21g$$

A1

3

(c) Resolve vertically $T_A + T_B = 40g$

M1

$$T_A = 19g \text{ or } 186 \text{ N}$$

A1

2

(d) Gravitational force acts through mid point of the rod

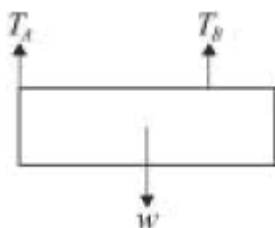
E1

1

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Q7.

(a)



Arrows + labels, w in centre

B1

1

- (b) M(A) $0.4W = 0.6T_B$
Moments equation

M1

$$T_B = \frac{2W}{3}$$

Accept 2 dp for each A1

A1

Res $\uparrow$ or M(B) $T_A = \frac{W}{3}$

M1A1

4

- (c) Lamina is uniform $\Rightarrow$ weight acts at centre

B1

1

[6]

Q8.

Moments about A:
OE

$$6S = 4 \times 30g$$

M1A1

$$S = 20g \text{ N or } 196 \text{ N}$$

A1

Resolving vertically: $R + S = 30g$

M1

$$R = 10g \text{ N or } 98 \text{ N}$$

SC3 20 N; 10 N

A1

5

[5]

Q9.

- (a) Centre of mass of rod is 3 m from river bank
Use of centre of mass is centre of rod

B1

Taking moments about A , edge of bank:
 $3 \times 15 = 50x$

M1

$$x = 0.9$$

A1

Or resolve $R = 65g$ B1
Moments about any point (correct) M1
 0.9 A1

3

- (b) Taking moments about A :
 $50 \times 2 = 15 \times 3 + m \times 8$
M1 3 terms, 2 correct

M1A1

$$55 = 8m$$

A1

$$m = 6\frac{7}{8}$$

$$\text{Mass is } 6\frac{7}{8} \text{ kg}$$

Accept 6.88 and 6.87

A1

4

- (c) Centre of mass of rod is 3 m from river bank
Centre of mass is at centre of rod

E1

1

- (d) eg Woman is a particle
 The mass is a particle
 The plank is a rigid rod

E1

1

[9]

Q10.

$$5T_A = 20 \times 9.8 \times 1.5$$

Moment equation.

M1

Correct equation

A1

$$T_A = \frac{20 \times 9.8 \times 1.5}{5} = 58.8 \text{ N}$$

Correct tension

A1

Vertical equation with T or moments equation.

M1

$$T + 58.8 = 20 \times 9.8$$

Correct equation

A1

$$T = 137.2 \text{ N}$$

Correct tension

A1

[6]

Q11.

(a)



correct force diagram, with labels and arrows.

B1

1

(b) $2T = 0.5 \times 98$

moment equation

M1

$$T = 24.5 \text{ N}$$

correct equation

A1

AG

*correct positive value for the tension from
correct working*

A1

3

(c) (i) $2 \times 2 \times 24.5 = 3 \times 9.8 \times m + 0.5 \times 98$
tension doubled

B1

moment equation

M1

$$m = \frac{98 - 49}{3 \times 9.8} = \frac{5}{3} = 1.67 \text{ kg (to 3 sf)}$$

correct equation

A1

correct mass

A1

4

OR

$$2 \times 24.5 = 3 \times 9.8m$$

for equation

(M1A1)

$$m = \frac{49}{29.4} = \frac{5}{3} = 1.67 \text{ kg}$$

for finding m

(M1A1)

(ii) $R = 24.5 \times 2 + 98 + \frac{5}{3} \times 9.8 = 163 \text{ N}$

considering vertical equilibrium with 3 terms

M1

correct equation

A1

correct reaction

A1

3

must be consistent with 3(c)(i) if awarding

accuracy marks

- (d) This allows the centre of mass to be placed at the centre of the rod for the moment calculations.

correct explanation

B1

1

[12]

Q12.

- (a) (i) $10 \times 9.8 \times 0.5 = 2T$

Moment about pivot with 2 terms

correct revision from correct working

M1

$$T = 24.5$$

Correct moment equation

A1 A1

3

- (ii) $10 \times 9.8 \times 0.5 + 40 \times 9.8 \times 3 = 2T$

Moments about pivot with 3 terms

Correct tension from correct working

Correct moment equation

M1 A1

$$T = 613 \text{ (to 3 sf)}$$

Correct tension from correct working

A1

3

- (b) No change, as the ratios of the distances from the pivot would be the same

No

Reason

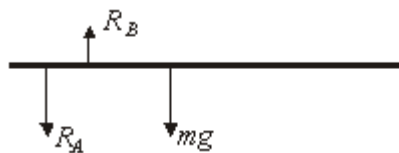
B1 B1

2

[8]

Q13.

- (a)



Correct force diagram

B1

1

(b) $0.8R_A = 0.5 \times 40 \times 9.8$

Moment equation to find R_A

M1

Correct equation

A1

$$R_A = \frac{0.5 \times 40 \times 9.8}{0.8} = 245 \text{ N}$$

ag *Correct reaction from correct working*

A1

3

(c) $R_B = 245 + 40 \times 9.8 = 637 \text{ N}$

Use of equilibrium to form an equation

M1

Correct reaction

A1

2

(d) $0.8R_A = 0.5 \times 40 \times 9.8 + 3 \times 5 \times 9.8$

Four term moment equation

M1

Correct equation

A1

$$R_A = \frac{0.5 \times 40 \times 9.8 + 3 \times 5 \times 9.8}{0.8}$$

Finding reaction

M1

$= 429 \text{ N (to 3sf)}$

Correct reaction

A1

$$R_B = 429 + 40 \times 9.8 + 5 \times 9.8$$

Equation to find other reaction

M1

$$= 870 \text{ N (to sf)}$$

Correct reaction

A1

6

[12]

Q14.

$$4R_B = 2.2 \text{ m}$$

Moments about A or B

Correct equation

Correct Reaction force

M1 A1

$$R_B = 216 \text{ N (to 3sf)}$$

A1

$$R_A + R_B = 40 \times 9.8$$

Moments about other end or resolve vertically

M1

$$R_A = 176 \text{ N (to 3 sf)}$$

Other reaction correct

A1

[5]